

From phonetic cradle to lexical grave



Demystifying the ‘pan-generational forces’ driving the life cycle of phonological change

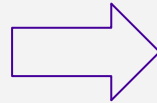
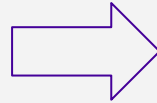
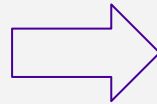
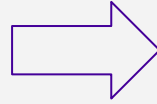
Ranjan Sen, University of Sheffield

Edinburgh Theoretical Historical Linguistics Seminar, 16-17 April 2025



Aren't we talking about the same thing?

- Historical linguists explore
 - As sound changes
- Synchronic phonologists explore
 - As 'rules' of the grammar
- Phoneticians explore
 - As connected speech processes
- Morphologists explore
 - As morphological operations
- The lexicon shows the same patterns
 - *Limp, Lint, Link; *Limk, *Linp*



Assimilation

Lenition

Insertion

Deletion

Vowel movements

Etc.

Sound changes have a life

- Sound changes have a chronological span
- OE sk > ʃ *scip* ‘ship’
- c. 1000: OE borrowed Latin *scrīnium* borrowed > *scrīn* ‘shrine’
- c. 1330: ME borrowed Old Norse *skyrta* ‘skirt’ - no longer active

- But (just like humans) sound changes behave differently when they’re younger vs. when they’re older
- So phonetics, phonology, morphology, and the lexicon all play a role at stages in the life of a single sound change

- Let us therefore investigate:

The life cycle of a phonological
process (synchronic focus)

The life cycle of a sound change
(diachronic focus)



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Structure of the session

- What are the stages of the life cycle?
 - Problems 1-2
- Interlude: A history lesson
- Insights of the life-cycle model
 - (1) The Neogrammarian controversy
 - (2) Rule aging (Problem 3)
 - (3) Dialect continua (Problems 4-5)
 - (4) Rule scattering
 - (5) Dual morphosyntactic conditioning
- What causes the life cycle?

What are the stages of the life cycle?



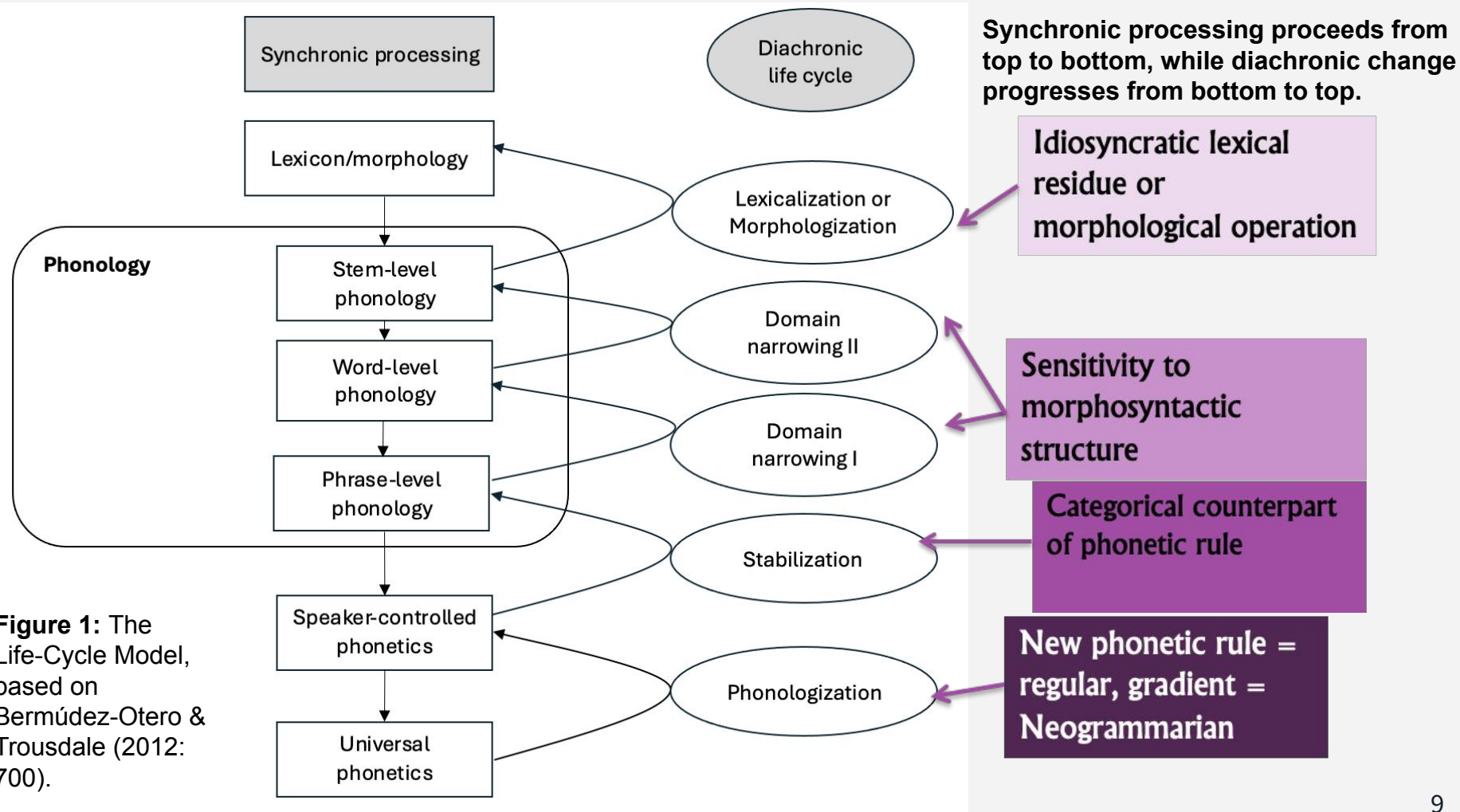
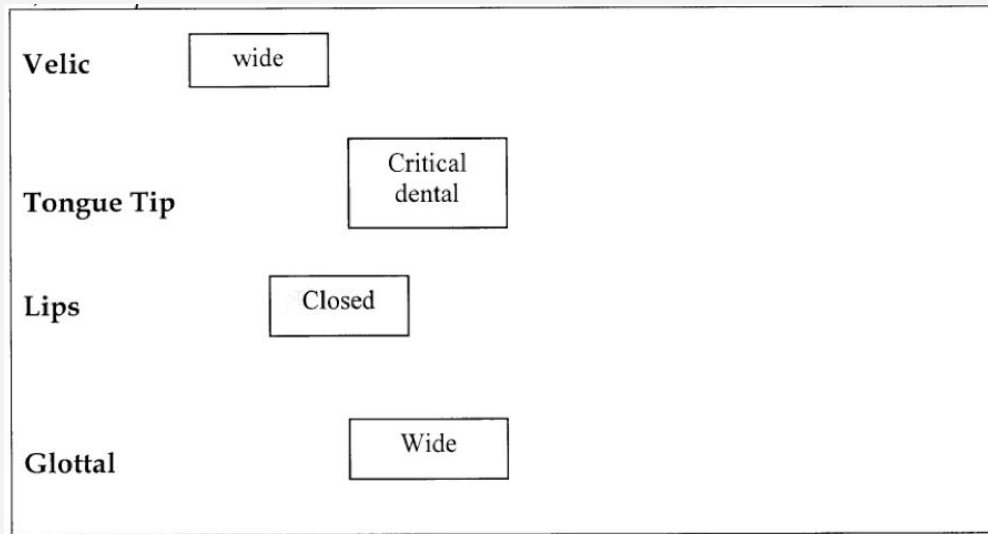


Figure 1: The Life-Cycle Model, based on Bermúdez-Otero & Trousdale (2012: 700).

Life cycle of a sound change: Stage 0

- Language-universal, automatic phonetics
- The result of physics (e.g. aerodynamics) and physiology, so **gradient**

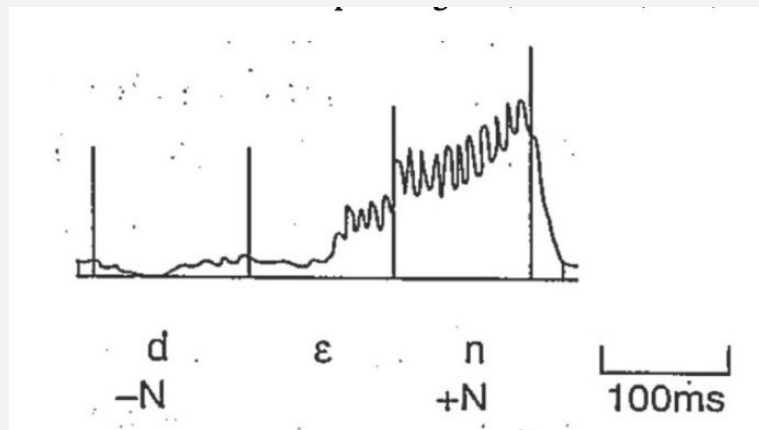


‘ham[p]ster’

Processes like this have their origin in physiological differences in the timing of the movement of speech organs

Stage 1: Speaker-controlled phonetics: Phonologization

- **Phonologization:** A mechanical phonetic effect is reinterpreted as a cognitively controlled, language-specific, gradient process of phonetic implementation



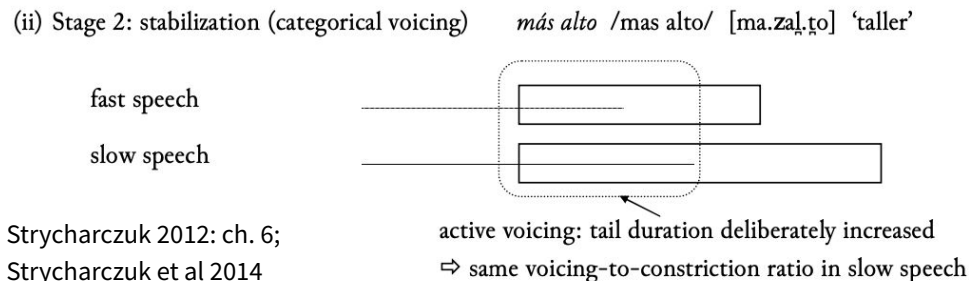
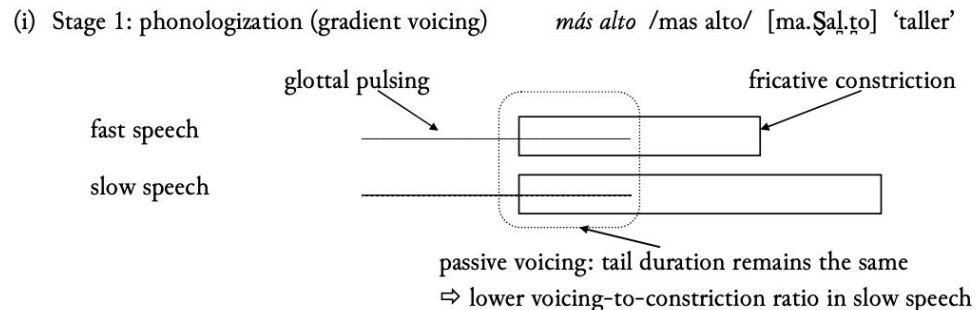
US English 'den' has nasal airflow beginning early in the vowel systematically.

Gradually rising contour indicates that the effect is gradient.

Cohn (1990, 1993)

Stage 2: Phrase-level rule: Stabilization

- **Stabilization:** A gradient process of phonetic implementation is reanalysed as a categorical phonological ‘rule’ applying across the board at the **phrase level**.
 - I.e. Insensitive to word boundaries; ‘phrase’ here refers to utterances.



Quito Spanish voicing of word-final prevocalic /s/.

Stage 2: Stabilized categorical voicing

Diagnosed by maintenance of the ratio of the /s/ that is voiced across different speech rates

Stage 3: Word-level rule: Domain narrowing I

- **Domain narrowing I:** The target and context for the rule must now be within a grammatical word; the domain of the rule **narrows** from ‘utterance’ to ‘word’
- **Dutch coda devoicing:** devoice obstruents in coda position
- But at which level? **Resyllabification** in *de held is* ‘the hero is’:
 - Word-level /d/ is in the coda: /də/ /hɛld/ /ɪs/
 - Phrase-level /d/ is in the onset: /də.hɛl.dɪs/
- Word-level! /hɛld/ → [hɛlt]

Stage 4: Stem-level rule: Domain narrowing II

- **Domain narrowing II:** The target and context for the rule must now be within a sublexical domain, e.g. stem; the rule **narrows** from ‘word’ to ‘stem’
- Some affixation occurs at the stem level
 - e.g. English ‘level one’ affixes (e.g. those of Latinate origin)
- **English stress assignment:** stem-level affixes cause stress shift; word-level affixes do not
- Stem level affixation: *ó*ri^gin → *orí*gin-*al* → *orì*gin-*ál*-ity
- Word level affixation: *mír*th → *mír*th-ful → *mír*th-ful-ness

- Stress assignment is at the stem level

Stage 4: Stem-level rule: Domain narrowing II

- **What about Dutch coda devoicing?**
- *held+in* ‘heroine’
- Stem-level /d/ is in the coda: /hɛld.-ɪn/
- Word-level /d/ is in the onset: /hɛl.dɪn/
- Surface form is [hɛl.dɪn] without devoicing
- So devoicing is indeed (as concluded earlier) at the word level (stage 3)

Domain narrowing in action: English /g/ deletion

(Bermúdez-Otero & Trousdale 2012)

James Elphinston: 18th cent. orthoepist

		<i>Utterance - final</i>	<i>Before C-initial word</i>	<i>Before V-initial word</i>	<i>Suffixed</i>	<i>Mono- morphemic</i>	<i>Time/ source</i>
		<i>sing </i>	<i>sing tunes</i>	<i>sing it</i>	<i>sing-er</i>	<i>finger</i>	
Stages 0-1	Phonetics	ŋg?	ŋg?	ŋg	ŋg	ŋg	EME
Stage 2	PL	ŋ (PL coda)	ŋ (PL coda)	ŋg	ŋg	ŋg	Elphinston (‘solemn’)
Stage 3	WL	ŋ	ŋ	ŋ (WL coda)	ŋg	ŋg	Elphinston (informal)
Stage 4	SL	ŋ	ŋ	ŋ	ŋ (SL coda)	ŋg	PDE

Stage 5(i): Lexical Storage: Lexicalization

- **Lexicalization:** The output of the rule becomes the underlying representation stored in the mental lexicon; the rule may cease to be operational (rule death)
- **English ‘umlauted’ plurals**
- Pre-Old English fronting rule before [i]:
- PDE: lexically stored irregular plural
 - Though pattern formation in the lexicon may play a part in processing: *feet, geese, teeth*

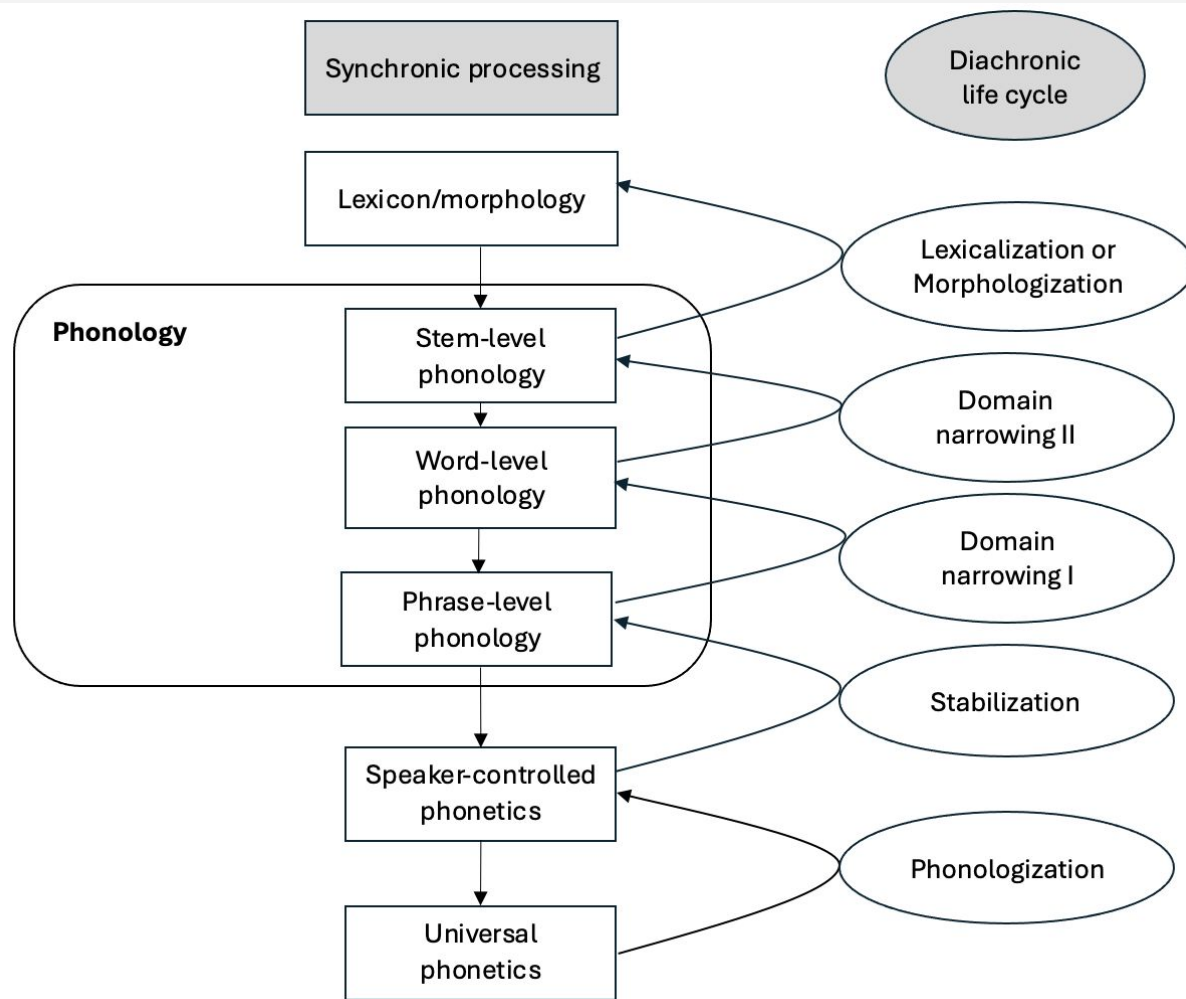
	Singular	Plural
UR	/fo:t/	/fo:t-iz/
SF	[fo:t]	[fø:tiz]
	‘foot’	‘feet’

Stage 5(ii): Morphological rule: Morphologization

- **Morphologization:** The phonological rule is reanalysed as a morphological operation of exponence; again, the phonological rule is no longer operational

Singular	Plural
Bruder	Brüder
'brother'	'brothers'

- **German umlauted plurals**
- Not a regular morphological process, but
- Now indicates plurality whereas it was once an assimilatory non-local vowel fronting before high, front vowels
- Underwent the full life cycle

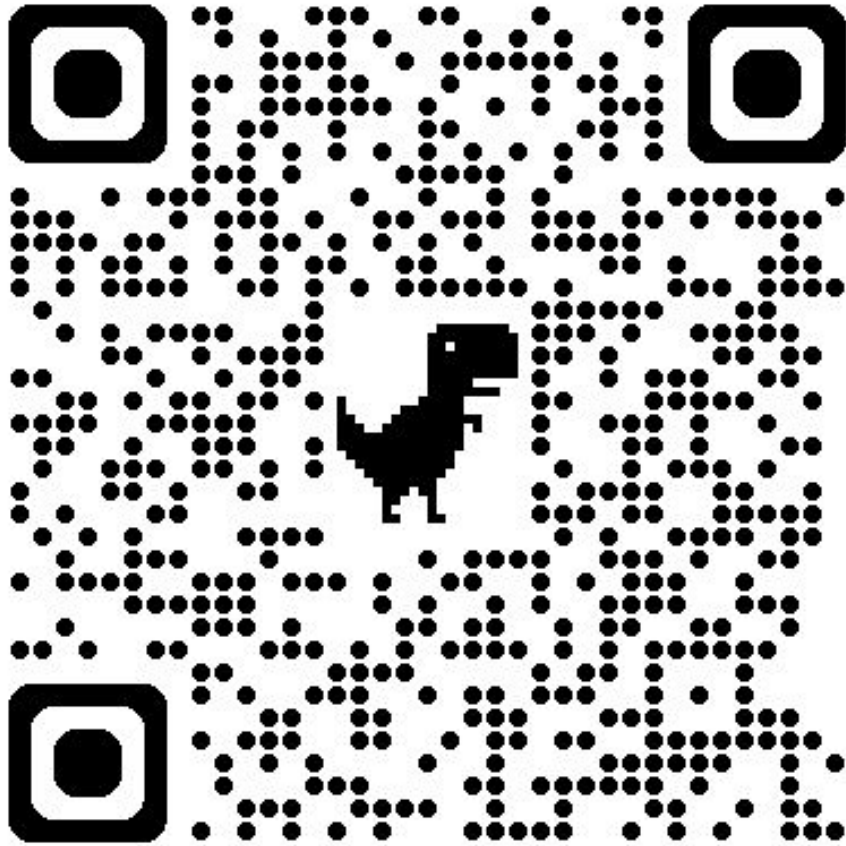


A sound change from its phonetic ‘cradle’ to its lexical (or morphological) ‘grave’

Phonetics → Phonology → Lexicon/Morphology

‘It should be as inconceivable for phonetic, phonological, and morphological research to proceed in ignorance of this life cycle as it is for research into morphology, syntax, semantics, and pragmatics to ignore the facts of grammaticalization’

(Bermúdez-Otero & Trousdale 2012)



Life Cycle Data Sheet

Problem (1): Gorgia Toscana (Ramsammy 2017)

- ‘GT is a variable lenition phenomenon that causes intervocalic stops to spirantise, either word internally or across word boundaries. It also applies to post-vocalic stops that are immediately followed by glide or a liquid’.
- *Libro pericoloso* ‘dangerous book’
/libro#perikoloso/ → Standard Italian [libroperikolozo]
Tuscan Italian [liβroφerixolozo]
- *La capra* ‘the goat’
/la#kapra/ → Standard Italian [lakapra]
Tuscan Italian [laxaφra]
- **At which phonological level is Gorgia Toscana operating?**

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Problem (2): Latin rhotacism (Roberts 2012)

Look at the data sheet, Problem (2): Guidance questions:

Section A: What happens to /s/ and in which environment? With the information that it does not occur across word boundaries, at which phonological level does this process appear to be occurring?

Section B: Does the process occur here? See the consonant following the /s/ for a potential explanation. Why might it not be an issue for A1 (see what happens to that form only later)?

Section C: Only forms in the later classical Latin are given. Why is the process not occurring here? At which phonological level would the process be occurring for your explanation to hold?

Section D: Has the process occurred here? If so, at which phonological level would it have to have occurred? Why are the forms in D different to those in C?

Section E: Does the process occur here? Explain the reason. Is the process still phonologically active in classical Latin? Where in the life cycle is it likely to be here?

Section F: Are these loanwords undergoing the process? Explain your answer.

Section G: How has the alternation changed between pre-classical and classical Latin? With reference to the life cycle, what has happened for this change to occur? What are the underlying representations of the two stems?

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Latin Rhotacism: Section A

		Reconstructed	Pre-classical	Classical	Translation	Notes
A	1	*sweso:r	soro:r	soro:r (later soror)	'sister'	
	2	*korpos ~ *korpos-is	corpus ~ corporis	corpus ~ corporis	'body'	nom ~ gen
	3	*ru:s ~ *ru:s-is	ru:s ~ ru:ris	ru:s ~ ru:ris	'countryside'	nom ~ gen
	4	*flo:s ~ *flo:s-is	flo:s ~ flo:ris	flo:s ~ flo:ris	'flower'	nom ~ gen

- Intervocalic rhotacism: $s \rightarrow r / V_V$
- These data show a word-level rule
- The /s/ is not intervocalic in the nominative (stem alone) *corpus* but it is in the genitive (stem + suffix) *korpos-is > *corporis*
- Section B illustrates a constraint: rhotacism fails when the /s/ is separated by another /r/ by only a short vowel: *miser*, *caesaries*

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Latin Rhotacism: Sections C and D

C	7	*de:-salio:	de:silio:	'I jump down'	cf. salio: 'I jump'
	8	*de:-sino:	desino:	'I desist'	cf. sino: 'I allow'
	9	*re-seko:	reseco:	'I cut back'	cf. seco: 'I cut'
	10	*ni:-si:	ni:si:	'unless'	cf. si: 'if'

- These data suggest that the rule narrowed to the stem level
- The /s/ is stem-initial, so not intervocalic in the stem alone: *sal-*, *sin-*

D	11	*dis-emo:	dirimo:	'I take apart'	Connection to prefix <i>dis-</i> lost by classical Latin
	12	*dis-(h)abeo:	diribeo:	'I distribute'	Connection to prefix <i>dis-</i> lost by classical Latin

- These data suggest that the rule originally applied at the word level in these forms
- After the rule narrowed to the stem level, rhotacism persisted as they were treated as stems, e.g. *dirim-*

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Latin Rhotacism: Sections E and F

E	13	*kaussa	caussa	causa	‘cause’	
	14	*wi:ssos	vi:ssus	vi:sus	‘seen’	
F	15			basis; cisium	‘pedestal’; ‘cabriolet’	Loanwords borrowed in or after classical Latin

- Section E: Degemination of /ss/ after long vowels/diphthongs occurs in classical Latin. Does this lead to new short /s/ being rhotacized? **NO**
- Why not? The process has been **lexicalized** and is not longer an operational phonological rule
- Section F: This is confirmed by its failure in loanwords borrowed at this time:
basis, cisium

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Latin Rhotacism: Section G

G	16	*hono:s ~ *hono:s-is	hono:s ~ hono:ris	honor ~ hono:ris	'honour'	nom ~ gen
	17	*arbo:s ~ *arbo:s-is	arbo:s ~ arbo:ris	arbor ~ arbo:ris	'tree'	nom ~ gen

- Section G: nominative *hono:s* > *honor*
- But /s/ is not intervocalic here
- And we just concluded that rhotacism was no longer active
- This is what is traditionally described as analogical levelling:
 - Stem *honor-* in the oblique cases
 - *honor* replaces nominative (only) stem variant *hono:s*
- In the life cycle, we account for this as **lexical structuring**, i.e. the stored UR of the stem morpheme now has final /-r/
- Not /-s/ plus active rhotacism in the oblique cases

Interlude:

A history lesson



A bit of (historical) history: Amphichrony

- The life cycle - an internal trajectory over the course of which sound changes become increasingly sensitive to grammatical information - has been noticed since the beginnings of Structuralist Linguistics (e.g. Kruszewski 1881; Baudouin de Courtenay 1895)
- *Pace* the ‘Saussurian firewall’ (Saussure 1916; Kiparsky 2015), diachrony therefore interfaces with synchrony, as diachronic stages are synchronic states
 - Stage 1: Synchronic phonetic rules
 - Stages 2-4: Synchronic phonological rules
 - Stage 5: Synchronic morphology or lexical patterns
- This approach can therefore be termed ‘amphichronic’

A bit of (historical) history: Stratal phonology

- The life cycle model (LCM) makes use of morphosyntactically defined derivational levels in phonology: stem, word, and phrase
- Builds on the insights of Lexical Phonology and Morphology (LPM; Kiparsky 1982)
 - Lexical rules, sensitive to word-formation and boundaries
 - Cyclic (recursive) rules after stem-forming
 - Post-cyclic (one time) rules on inflected words
 - Post-lexical rules, can cross boundaries
- Soon became clear that LPM could model the life cycle (Kiparsky 1988; Harris 1989; McMahon 1991, 2000): phonetics → post-lexical → post-cyclic → cyclic → UR
 - But LPM had irreconcilable difficulties: <http://www.bermudez-otero.com/lexphon.pdf>
- Current successor is Stratal Phonology, mainly applied using Stratal Optimality Theory (N.B. traditional OT is parallel and has no levels)

Insights of the LCM

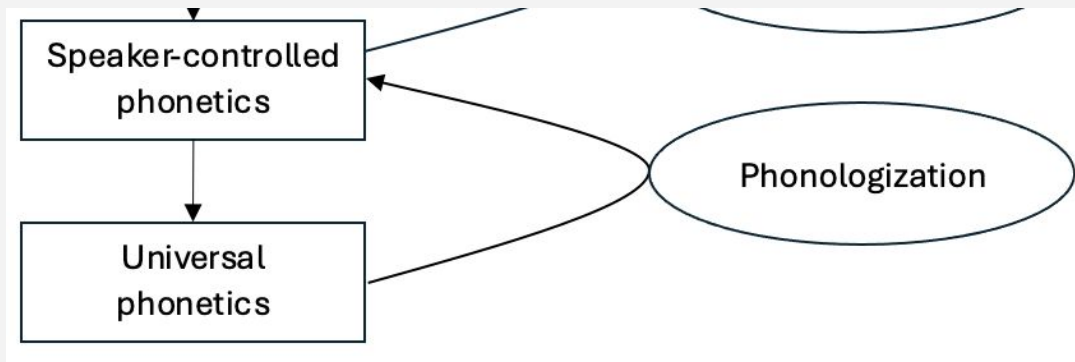


(1) The Neogrammarian Controversy

- Long-standing problem: **Is sound change regular?**
 - I.e. Does it affect all words containing the phonological context at the same time, or does it operate word by word?
 - Concomitantly, does it proceed phonetically gradually, or abruptly, i.e. categorical replacement of one sound for another?
- **Neogrammarian Hypothesis** (19th cent. Leipzig school): regular and gradient
- **Lexical diffusion** (Wang 1969; Chen and Wang 1975): one word at a time, sweeping through the lexicon by categorically replacing target sounds in a phonetically abrupt way
- Labov (1981) concluded that both were genuine, but separate types of change

What does the LCP predict: Stage 1

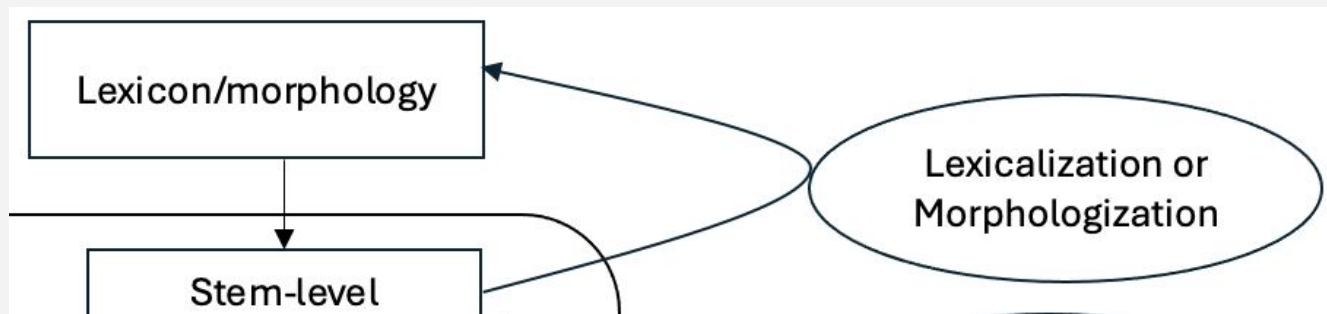
- Stage 1, Speaker-controlled phonetics after phonologization:
Articulatory targets are assigned gradiently without interfacing with the lexicon, so all words where the rule's context applies are affected = phonetic rule
- This is regular Neogrammarian sound change
 - E.g. Labov (2010) u:-fronting; Fruehwald (2012) aɪ-raising



Phon: gradual
Lex: abrupt

What does the LCP predict: Stage 5

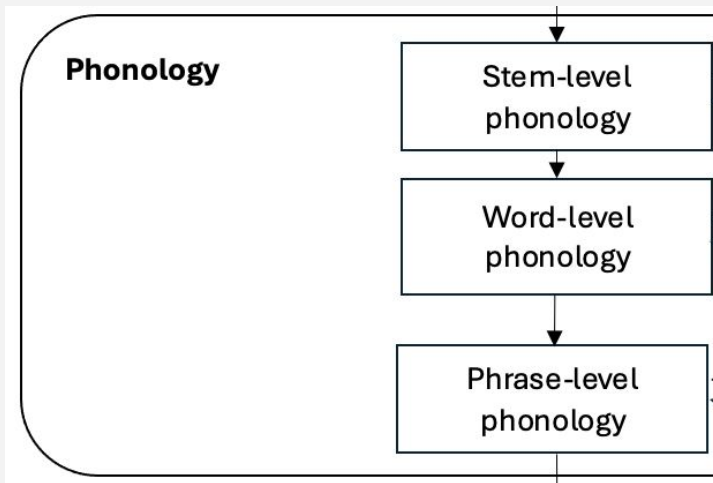
- Stage 5, Lexical storage by lexicalization:
The stored URs of individual lexical items are restructured to be the post-rule form; this can occur word by word depending on e.g. frequency
- This is classical Lexical Diffusion



Phon: abrupt
Lex: gradual

What does the LCP predict: Stages 2-4

- Stages 2-4, Phonological rules:
Categorical processing whereby inputs are altered in specific environments
- As amply attested



Phon: abrupt
Lex: abrupt

(2) Rule aging

- Correlation between relative ages of rules and their positions in the grammar
 - Older rules apply in higher strata
- Lenition trajectories: diachronically, predict milder lenitions before more extreme ones
 - Older rule: mild reduction
 - Newer rule: more severe reduction
- On the basis of the life cycle, we predict:
 - Milder reduction at higher level
 - More severe reduction at lower level
- Prediction borne out!
 - We should be on the lookout for this in our work!

Problem (3) English /l/ darkening and vocalization

Bermúdez-Otero & Trousdale (2012)

	<i>see Lynn</i>	<i>seal in</i>	<i>seal bins</i>
WL	si: . lɪn	si:ɫ . ɪn	si:ɫ . bɪnz
PL	si: . lɪn	si:ɫ ɪn	si:ʊ . bɪnz

- ɫ = dark /l/, ʊ = vocalized /l/
- Look at the Word Level. What happens to /l/s here and what is context for it?
- Look at the Phrase Level. What happens to /l/s here and in what context?
- Which process is a more extreme change: the WL one or the PL one?
- Is the milder process at a higher level, as predicted?

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English /l/ darkening and vocalization

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	<i>see Lynn</i>	<i>seal in</i>	<i>seal bins</i>
WL	si: . lɪn	si:t . ɪn	si:t . bɪnz
PL	si: . lɪn	si:t̚ ɪn	si:ʊ . bɪnz

- Look at the Word Level. What happens to /l/s here and what is context for it?
 - **/l/-darkening in syllable codas at the Word Level (Ask: is /l/ in coda position within a word?)**
- Look at the Phrase Level. What happens to /l/s here and in what context?
 - **/l/-vocalization in syllable codas at the Phrase Level (Ask: is /l/ in coda position within an utterance? Recall resyllabification)**
- Which process is a more extreme change: the WL one or the PL one?
 - **/l/-vocalization is more extreme than /l/-darkening (tongue tip gesture lost)**
- Is the milder process at a higher level, as predicted? **YES**

Rule aging: rule generalization

- Not just lenitions, but any rule whose scope becomes more general over time can be predicted to obey this pattern in a synchronic grammar:
 - More specific rule: higher level
 - More general rule: lower level
- Why do rules generalize? (‘phonetic analogy’, Schuchardt 1885)
- One reason (but perhaps not the only one) is that:
 - Change often begins in specific environment with phonetically favourable conditions
 - Progressively spreads to more general contexts based on scalar phonetic favourability
 - E.g. Caraqueño final nasal velarization rates: __ Dor > __Lab > __Cor (Ramsammy 2015)
- This may result in a **dialect continuum** of more and less advanced varieties
 - OHG (af)frication: V__(north) > C__ > #__(south) (Bermúdez-Otero 2015)

(3) Dialect continuum: Space and Time

- A life-cycle-based dialect continuum is predicted according to the LCM
 - A rule is at a lower level in more conservative dialects
 - A rule is at a higher level in more innovative dialects
- ‘Synchronic phonology of one variety may reflect a historical stage of a more advanced dialect or a potential future stage of a more conservative dialect’ (Ramsammy 2015)
- **Problem (4): Velarizing Spanish**

		Non-velarising Spanish	Peninsular Velarizing Spanish (PVS)	Galician (functional vocabulary only)	
A	1	[pan]	[paŋ]		‘bread’
	2	[pan-aθ-o]	[panaθo]		‘bread.AUG’
	3	[pan aθimo]	[paŋ aθimo]		‘unleavened bread’
	4	[kome pan]	[kome paŋ]		‘eats bread’
B	5		[uŋ] ~ [una]	[uŋ] ~ [uŋa]	‘a’ M ~ F
	6		[alxuŋ] ~ [alxuna]	[alxuŋ] ~ [alxuŋa]	‘some’ M ~ F
	7		[niŋguŋ] ~ [niŋguna]	[niŋguŋ] ~ [niŋguŋa]	‘no’ M ~ F
	8		[de uŋ] ~ [de una]	[duŋ] ~ [duŋa]	‘of a’ M ~ F

Guidance questions

1. Section A: Where is /n/ realised as velar [ŋ] in Peninsular Velarizing Spanish? At which phonological level does this rule operate?
2. Section B: Compare the PVS alternations with those in Galician (functional vocabulary). Does PVS behave according to your conclusion in Section A? Does Galician behave differently? At what level is the rule operating here? Could it have progressed even further along the life cycle?

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Dialects: life cycle AND rule generalization

- So a dialect continuum can reflect rule age in two ways:
 - Specific↔general versions of a rule appear in conservative↔innovative dialects
 - A rule is lower↔higher in the life cycle in conservative↔innovative dialects
- But what is further predicted is both the life cycle and rule generalization operating together within a dialect continuum:
 - A sound change is phonologized and stabilized environment by environment
 - Each rule goes through the life cycle independently
 - The change is at higher levels in more innovative dialects and lower levels in more conservative
- We should be looking for this everywhere!
- **Problem (5): Swiss German umlauting and lowering**

			Translation	St Galler Rheintal	Schaff- hausen	Kesswil
A	1	/ops/	'fruit'	[ops]	[ops]	[ops]
	2	/bokx/	'ram'	[bokx]	[bokx]	[bokx]
	3	/holts/	'wood'	[holts]	[holts]	[holts]
	4	/wonə/	'live'	[wonə]	[wonə]	[wonə]
B	5	/forə/	'Scots pine'	[forə]	[forə]	[forə]
	6	/horn/	'horn'	[horn]	[horn]	[horn]
C	7	/boge/ ~ /boge-PL/	'bow/bows'	[boge] ~ [bøge]	[boge] ~ [bøge]	[boge] ~ [bøge]
	8	/torn/ ~ /torn-PL/	'thorn/thorns'	[torn] ~ [tœrn]	[torn] ~ [tœrn]	[torn] ~ [tœrn]
	9	/xorb/ ~ /xorb-li/	'basket/basket.DIM'	[xorb] ~ [xœrbli]	[xorb] ~ [xœrbli]	[xorb] ~ [xœrbli]
D	10	/bodə/ ~ /bodə-PL/	'floor/floors'	[bodə] ~ [bødə]	[bodə] ~ [bødə]	[bodə] ~ [bødə]
	11	/trottə/ ~ /trottə-li/	'sidewalk/sidewalk. DIM'	[trottə] ~ [trøtli]	[trottə] ~ [trøtli]	[trottə] ~ [trøtli]

- Section A: Are there any phonological processes in play? Take note of the post-vocalic consonants in this section.
- Section B: Are there any processes occurring? How might the process be described (see IPA key above)? What is the triggering environment? Give the process a name, treating the target input as specifically the one seen in the data.
- Section C: See C7. How is the plural formed? The plural morpheme can be interpreted as a 'floating feature' (i.e. just a phonological feature, nothing else). What could this feature be based on how the plural is formed? Give the process a name.
- See C8-9. How do the singulars/monomorphemic forms differ from C7 and is this difference expected based on your previous conclusions?
- See C9. Why is the process you have formulated for Section C occurring here?
- How are the derived forms in C8-9 different from C7? Is this expected based on your conclusions? Which process would need to come earlier/at a higher level in the phonology?
- See Section D. Does St Galler Rheintal behave as expected?
- What are Schaffhausen and Kesswil doing in the monomorphemic forms that cannot be accounted for by the processes so far? Formulate this as a separate process, taking note of the triggering environment following the vowel, and keeping the target input specific to the data.
- What is Schaffhausen doing in the derived forms? Did you expect this? Which process is coming earlier in the phonology: the one you formulated in Section C or in D?
- What is Kesswil doing in the derived forms? Did you expect this? Which process is coming earlier in the phonology: the one you formulated in Section C or in D?
- Where in the phonology do your three processes occur? Hints: (a) Your Section B rule occurs within stems; (b) Your Section C rule occurs predominantly in inflected forms; (c) Your Section D rule must occur at a level above your Section C rule in the Kesswil dialect, but at the same or below in Schaffhausen.
- Does your Section D rule look similar to your Section B one? Is it more specific or more general? Is its phonological level higher or lower?
- Draw derivation tables showing how 'thorn/thorns' and 'floor/floors' are derived in the three dialects. Your tables should show URs at the top, SFs at the bottom, and the processes at appropriate levels (SL, WL, PL) in between.
- Which dialect do you think is the most conservative and which is the most innovative? What makes a dialect more innovative from a life-cycle perspective?

			Translation	St Galler Rheintal	Schaff- hausen	Kesswil
A	1	/ops/	‘fruit’	[ops]	[ops]	[ops]
	2	/bokx/	‘ram’	[bokx]	[bokx]	[bokx]
	3	/holts/	‘wood’	[holts]	[holts]	[holts]
	4	/wonə/	‘live’	[wonə]	[wonə]	[wonə]

Section A: Are there any phonological processes in play?
Take note of the post-vocalic consonants in this section.

No processes in this section

/o/ is realised as [o]

Before [p, k, l, n]

Non-coronals and coronal sonorants

			Translation	St Galler Rheintal	Schaff- hausen	Kesswil
B	5	/fore/	‘Scots pine’	[f ^ɔ rə]	[f ^ɔ rə]	[f ^ɔ rə]
	6	/horn/	‘horn’	[h ^ɔ rn]	[h ^ɔ rn]	[h ^ɔ rn]

Section B: Are there any processes occurring? How might the process be described (see IPA key above)? What is the triggering environment? Give the process a name, treating the target input as specifically the one seen in the data.

Pre-/r/ /o/-lowering (to [ɔ])

In all these dialects

C	7	/bogə/ ~ /bogə-PL/	'bow/bows'	[bogə] ~ [bøgə]	[bogə] ~ [bøgə]	[bogə] ~ [bøgə]
	8	/torn/ ~ /torn-PL/	'thorn/thorns'	[tɔrn] ~ [tœrn]	[tɔrn] ~ [tœrn]	[tɔrn] ~ [tœrn]
	9	/xorb/ ~ /xorb-li/	'basket/basket .DIM'	[xɔrb] ~ [xœrbli]	[xɔrb] ~ [xœrbli]	[xɔrb] ~ [xœrbli]

Section C: See C7. How is the plural formed? The plural morpheme can be interpreted as a 'floating feature' (i.e. just a phonological feature, nothing else). What could this feature be based on how the plural is formed? Give the process a name.

Floating feature [-back] = Plural morpheme
Umlauting/fronting

See C8-9. How do the singulars/monomorphemic forms differ from C7 and is this difference expected based on your previous conclusions?

C8-9 have following /r/ so undergo expected /o/-lowering

See C9. Why is the process you have formulated for Section C occurring here?

Front vowel with [-back] in the diminutive suffix /li/

How are the derived forms in C8-9 different from C7? Is this expected based on your conclusions? Which process would need to come earlier/at a higher level in the phonology?

They have undergone pre-/r/ /o/-lowering *before* being umlauted/fronted:

/torn/ + [-back] → tɔrn + [-back] → [tœrn]

/xorb/+/li/ → xɔrbli → [xœrbli]

In all these dialects

			Translation	St Galler Rheintal	Schaffhausen	Kesswil
D	10	/bodə/ ~ /bodə-PL/	'floor/floors'	[bodə] ~ [bødə]	[b ^ɔ də] ~ [bødə]	[b ^ɔ də] ~ [b ^æ də]
	11	/trottə/ ~ /trottə-li/	'sidewalk/ sidewalk. DIM'	[trottə] ~ [trøtli]	[tr ^ɔ tte] ~ [trøtli]	[tr ^ɔ tte] ~ [tr ^æ tli]

Most conservative: St Galler Rheintal

- General lowering has not (yet) entered the grammar

Most innovative: Kesswil

- General lowering has not only entered the grammar, but also reached the stem level (so occurs derivationally earlier than umlauting)

See Section D. Does St Galler Rheintal behave as expected? **Yes**
What are Schaffhausen and Kesswil doing in the monomorphemic forms that cannot be accounted for by the processes so far?

/o/-lowering, but not before /r/

Formulate this as a separate process.

Pre-CORONAL /o/-lowering

What is Schaffhausen doing in the derived forms? Did you expect this? Which process is coming earlier in the phonology: the one you formulated in Section C or in D?

Umlauting o → ø

Before lowering, so umlauting prevents this: *o → ɔ → œ

This is the opposite order to pre-/r/ /o/-lowering

What is Kesswil doing in the derived forms? Did you expect this?
Which process is coming earlier in the phonology: the one you formulated in Section C or in D?

Lowering BEFORE umlauting, so precisely the opposite of Schaffhausen: o → ɔ → œ

This is the same order as pre-/r/ /o/-lowering

Where in the phonology do your three processes occur?

SL: Pre-/r/ /o/-lowering, Kesswil general lowering

WL: Umlauting, Schaffhausen general lowering

Does your Section D rule look similar to your Section B one? Is it more specific or more general? Higher or lower?

Pre-coronal lowering is more general than pre-/r/ lowering

Going through the life cycle separately and later

It is still at a lower level in Schaffhausen, as expected

(4) What does the LCM NOT predict

Phon: grad
Lex: grad

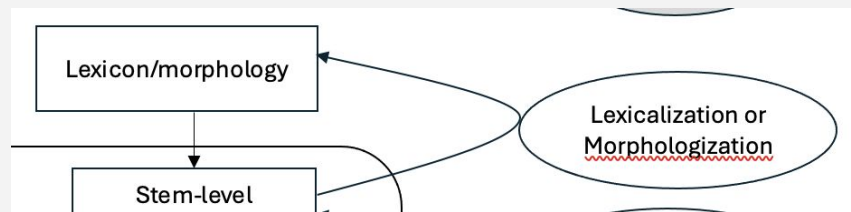
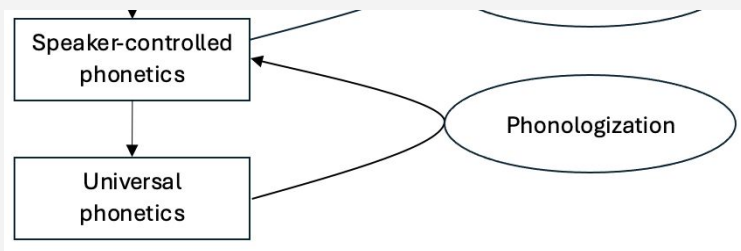
- There is nowhere in the architecture where a process can be both phonetically gradient and affect words one at a time, or be sensitive to any lexical/morphological information
- Yet such phenomena are claimed to occur (e.g. Bybee 2001; Phillips 2015)
- Is there a mechanism whereby gradience and categoricity can apply simultaneously?
- Rule scattering

(4) ‘Rule scattering’ (Bermúdez-Otero 2007, 2015)

- When rules rise, they don’t necessarily also leave the level they’ve vacated; the rule can operate at both/all levels to that stage = **rule scattering**
- This can explain the ‘illusion’ of morphological effects
- English /l/-darkening at increasingly higher rates in:
- light > gai-ly > Hayley > hail-y > mail it > bell (Sproat & Fujimura 1993; Boersma and Hayes 2001)
- Turton (2012): Categorical /l/-darkening rules at SL, WL, and PL: the more levels the rule goes through, the more likely the /l/ to be darkened
 - Darkening is near-obligatory at word and phrase levels, but only 78% of the time at the stem level

(4) Rule scattering: phonetics overlays phonology

- Crucially, this also occurs at the phonologization step, so a **gradient phonetic** rule can co-exist (and be overlaid on) its **categorical phonological** counterpart
 - Stem-level categorical palatalization (*confession*) and gradient phonetic palatalization (*press you*)
 - This is the case with Gorgia Toscana, seen as a PL rule earlier (Ramsammy 2017)
- A process can therefore have both categorical and gradient implementations simultaneously,
- predicting changes that appear to be both:
 - Phonetically gradual, and
 - Lexically gradual (item by item lexicalisation occurring)



(4) Rule scattering: phonetics overlays phonology

- We should be on the lookout for this! Work is ongoing:
- Bermúdez-Otero (2015): Philadelphia æ-tensing at (1) stem-level and (2) phonetic implementation, sensitive to similar factors
- losad (2025): Rule scattering explains repeated innovation (pertinacity) of Uralic secondary lengthening: persistent phonetic rule stabilises in different ways

Secondary lengthening	Fi	Ingr	Est	FiDial	Sámi
Phonetic rule	✓	☑	☑	☑	?
C > C ⁺ stops		✓	☑	☑	☑
C > C: stops			✓	✓	✓
C > C ⁺ all		✓		☑	☑
C > C: all				✓	✓

(5) Dual morphosyntactic conditioning

(Bermúdez-Otero & Luís 2009, Bermúdez-Otero 2011)

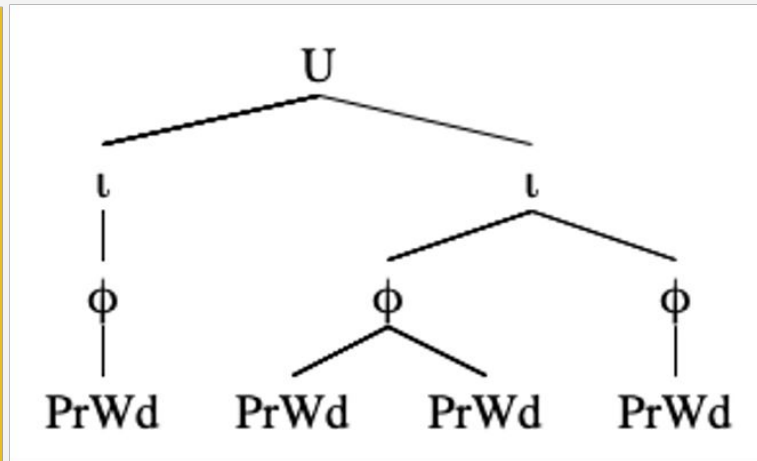
- Shortening sensitive not only to stratal computational procedure but also prosodic representational structure

Utterance

Intonational
Phrase

Phonological
Phrase

Prosodic Word



E.g. Match Theory (Selkirk 2011)

- Syllables can align to morphemes
- PrWds ω to GrWds
- Phonological Phrases ϕ to XPs
- Intonational Phrases ι to clauses

Prediction: morphosyntactic structure can condition a phonological process through both rule level (cyclically) and prosodic structure (representationally)

Latin iambic shortening (Sen, for-thcoming, -ever)

- LH → LL: A sequence of light syllable + heavy syllable is treated as two light syllables in early Latin
- There are four types of iambic shortening, but here are two key ones:
- PrWd iambic shortening: LH (́LL)
 - *homo:* (hó.mo) ‘human’
 - *potest* (pó.těst) ‘can’
- Cross-word iambic shortening: L#H... (L#L...)
 - *sed ostendere* (se.dös).(ten).(de.re) ‘but to show’
 - *hic est* (hi.cěst) ‘this is’
- Based on the latter (and other evidence), iambic shortening is clearly a phrase-level process in early Latin

The life cycle of Latin iambic shortening

- Early Latin verse (e.g. Plautus): **phrase-level**
 - Cross-word, with resyllabification
 - After WL stress assignment
- Early ‘popular’ verse inscriptions: **word-level**
 - Limited to single words of iambic shape
 - Transparent interaction with word-level stress assignment
 - No more cross-word shortenings!
- Later republican/imperial verse: word-level with further restrictions
 - Only shortening of long V, not closed syllable
- Imperial Latin: **lexicalized**
 - bene ‘well’, male ‘badly’, modo ‘only’, cave ‘beware’, ego ‘I’, and mihi tibi sibi ‘me/you/him (dat.)’
 - Only long-vowelled forms with PrWd shortening underwent lexicalisation = result of the internal history of the change
 - High token frequency forms - lexical diffusion
- The full life cycle (nearly!)

Prosodic conditions of iambic shortening

- In early Latin, shortening occurred **within phonological phrases** (ϕ), which were **not heads of their intonational phrases** (ι) in feet which were not heads of head PrWds
- I.e. in prosodically non-prominent positions

- Example: Plautus *Trinummus* 964
- *Vel trecentis. :: Haben tu id aurum quod accepisti a Charmide?*
- **Sycophant**: [You can] even [have] three hundred [words with me] ::
- **Charmides**: Have you that gold which you received from Charmides?

- NP with relative clause:
aurum quod accepisti a Charmide
- (Whole sentence is intonational phrase)

Prosodic conditions of iambic shortening

- Syntactic lexical phrases

[[aurum]_{NP} quod acce:pisti: a: [Charmide:]_{NP}]_{NP}

- Phonological phrases (Match XP)

[[aurum]_φ quod acce:pisti: a: [Chármide:]_φ]_φ

- Word-level foot structure (primary stresses)

[[(aú)rum]_φ (quód) (à)cè:(pís)ti: (á:) [(Chárr)mide:]_φ]_φ

- Phrase-level foot structure

consonant resyllabification, vowel elision, and **iambic shortening**

[[(aú)(rùm)]_φ (**quó.d_ăc**). (cè:). (pís). (ti: _á:) [(Chárr)(mìde:)]_φ]_φ

- **The shortening LH, a phrase-level foot, is not the head φ in the ι (Charmide), nor is it the head foot of the head PrWd (final or focused) in the φ**

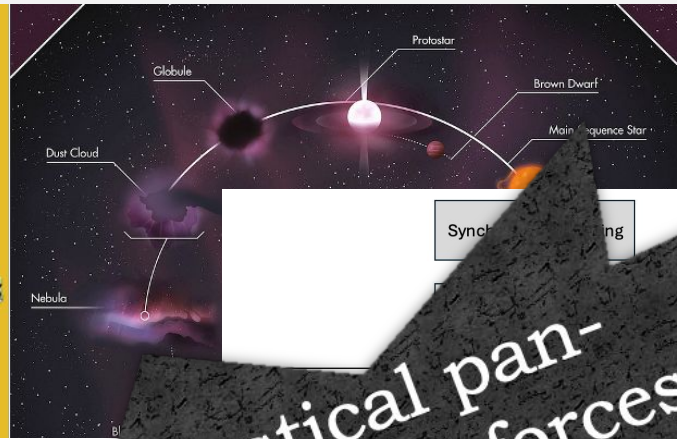
Dual morphosyntactic conditioning

- Phrase-level: utterance
- Prosody: φ-internal

**And finally:
What causes the
life cycle?**



Anise Swallowtail Life Cycle



The Twitter Life ©

Interact Level

"That's the dumbest thing I've ever seen!"
 "Woah, what are these cool things?"
 "Hmm, D'Arcy says it's cool."

Time Since Account Creation

Mystical pan-generational forces'!!
 (Hale et al. 2015)

Speaker-controlled phonetics
 Universal phonetics

Domain narrowing I

Stabilization

Phonologization

Unidirectionality: does life only go one way?

- Does ‘domain broadening’ occur (SL → WL → PL)?
 - N.B. Degrammaticalization in morphosyntax seems to exist (Norde 2009)
- Yes, but only in adult borrowing (Labov 2007: 369)
 - From: NYC stem-level short-æ tensing
 - To: New Orleans word-level short-æ tensing
 - Mis-internalization of opaque forms: SL as WL (Bermúdez-Otero & Trousdale 2012)

		<i>New York</i>	<i>New Orleans</i>
(Cardinal) Manning	WL SL Manning	[æ]	[æ:]
mann-ing (the pumps)	WL SL man ing	[æ:]	[æ:]

- ‘Unidirectionality has been found to break down in circumstances that favour higher rates of replication error than intergenerational transmission within a speech community’ (Bermúdez-Otero 2015)

Why so rare in intergenerational transmission?

- dada dad-a dad a dad dad > PL dada dad-a dad a dat dat
 - 2 out of 3 word-final /d/ devoiced... so generalized:
- > WL dada dad-a dat a dat dat
 - 3 out of 4 stem-final /d/ devoiced... so generalized:
- > SL dada dat-a dat a dat dat
- Every instance of /dad/ now [dat]
 - (could lexicalize as /dat/, but let's keep at SL)
- **But then...**
- SL dada dat-a dat a dat dat
 - 4 out of 5 non-word-initial /d/ devoiced... generalize?
- > WL dat a dat-a dat a dat dat
 - 5 out of 9 non-phrase initial /d/ devoiced...?! (token frequency)
- > PL data tat-a tat a tat tat??
- Rule generalization plus **domain broadening???**

Psycholinguistic roots of the life cycle

- Crucially, only the domain narrowing, not the broadening, involved morpheme alternation
- Hypothetical broadening relied on gross phonotactic generalizations
- **Anti-Alternation Bias:** Bias in acquisition (at 18+months) to reduce morpheme alternation (Bermúdez-Otero 2019)
- English /g/-deletion: $g \rightarrow \emptyset / \eta _ C, \#$ creates $g \sim \emptyset$ alternation
- Over-applying to smaller domains... reduces alternation
- Learner generalizes pattern if **frequency** of post-change alternant sufficiently high vs. unchanged alternant, as seems to have been the case in English (Bermúdez-Otero & Trousdale 2012)



How to test? Adult Artificial Language Learning?

- Can inaccurate learning of a fabricated language in the laboratory replicate the roots of domain narrowing?
- Test the acquisition of a Phrase-Level (PL) rule
 - Exposure to the more frequent alternant in training
- Do participants generalise to a Word-Level (WL) rule?
 - Reducing morpheme alternation, over-applying rule
- Let's have a (very rough) go... 😂

Cheat sheet (done by training phase): Stimuli

- Noun
- [pab] ~ [pav] ‘horse’
- Two others in actual experiment

- Suffixes
- [ib] ~ [iv] plural
- [deb] ~ [dev] feminine
- One more in actual experiment to manipulate input ratios

- Intransitive verbs
- /ama/ ‘run’
- /dumi/ ‘eat’
- One more in actual experiment to manipulate input ratios

Are you ready?

First, some training









Now for the test:
Choose which sentence is correct
for these two pictures: A or B



A



B



Your wooclap poll will be displayed here



Install the **Chrome** or
Firefox extension



Make sure you are in
presentation mode

wooclap



A



B

Your wooclap poll will be displayed here



Install the **Chrome** or
Firefox extension



Make sure you are in
presentation mode

wooclap

Did we overapply more in Test than Control?

- Rule per training data: syllable-final lenition: **b → v / __]σ @ the Phrase Level**
- Occurs when final or before C in a domain
 - /pabteb/## [pav.tev]##
- Fails when before V in a domain: resyllabified as onset
 - /pab-ib ama/ → [pa.bi.ba.ma] ‘the horses run’ - rule does not apply @ PL
 - /pab-teb dumi/ → [pav.tev.dumi] ‘the horse.fem eats’
- Test items:
- ‘The horses fly’ - Test condition (i.e. rule should not apply)
 - /pab-ib imu/ → [pa.bi.bi.mu] - rule does not apply @ PL
 - BUT → [pa.bi.v#i.mu] - **rule does apply @ WL - domain narrowing**
- ‘The horses sleep’ - Control condition (i.e. rule applies)
 - /pab-ib gimi/ → [pa.biv.gi.mi] - rule applies @ PL
 - Alternative → [pa.viv.gi.mi] - onset lenition (incorrect), unlikely domain narrowing to SL?

Final messages (and let's share our findings!)

- The life cycle is/should be (if correct) everywhere
 - The main way in which synchronic structure can explain diachronic phonology and vice versa
- Look out for domain narrowing (or indeed cases of broadening)
- Look out for milder/more specific rules being further along the life cycle than their more severe/more general counterparts
- Look out for life-cycle based dialect continua
- Look out for dual morphosyntactic conditioning
- The life cycle needn't rely on 'mystical pan-generational forces'
 - We can test the validity of the psycholinguistic underpinnings in the lab
- Historical linguistics at a complex crossroads: theoretical linguistics, philology, phonetics, psycholinguistics, sociolinguistics, language acquisition

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